Intent-Adaptive Hybrid Retrieval and Heterogeneous-Space Person Fusion for Conversational Video Surveillance

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***Abstract*—Form-driven alert editors force operators to hand-translate behavior into typed rules. Three contributions address conversational surveillance video. (C1) A *verified alert compiler* materializes fully-typed 9-tuple alert rules from free text with a deterministic keyword/regex materializer and a clarify/abstain validator; we report 91% rule-correctness in a separate deployment study of 60 utterances (LLM-backed materializer), and the release adds three reproducible probes: compositional held-out 9/9 vs. 51/51, oracle dialogue at 1.08 vs. 2.58 mean turns, and seeded fuzz with zero crashes, zero soundness violations and zero false materializations, plus a fixed adversarial round. (C2) A *heterogeneous-space person descriptor* fuses a SigLIP visual embedding ($\mathbb{R}^{768}$) with a MiniLM attribute-text embedding ($\mathbb{R}^{384}$) into one L2-normalized $\mathbb{R}^{1152}$ vector by weighted concatenation, not same-space addition; an ablation shows a 10.7-pp F_1 gain over the best unimodal index and 2.2-pp over rank-based late fusion (RRF); the released track is rebuilt on metadata-formed pseudo-tracks with threshold-free pair AUC as the primary metric. (C3) An *intent-adaptive routing rule* $\alpha(F, \iota, n_s, n_v)$ gives a closed-form per-query weight from filter F, intent ι and branch-size imbalance, plus hard overrides for counting and action queries (Eq. (5)); no fixed α attains both per-category optima (Lemma 1); a per-intent lookup reaches 0.822 vs. 0.828, so soft terms add $+0.006$ and overrides carry the effect, raising mean Precision@5 (macro-mean over five categories) on a 150-query benchmark from 0.67 (best fixed-α, α=0.3) to 0.83, while a score-confidence ablation is reported as a null on the released subset (three-way tie at 0.538; max-combination attains the zscore ceiling). They run in a nine-model pipeline at 2 FPS per camera, with 109 ms median FAISS lookup over 312k descriptors. The code paths behind the deployment numbers are exercised end-to-end by the supplementary artifact (https://github.com/sujalmh/intent-adaptive-hybrid-surveillance); the deployment per-query export was not preserved, so its CIs and p-values are not recomputable.**

***Index Terms*—Conversational Alert Authoring, Multimodal Person Re-ID, Heterogeneous Embedding Fusion, Hybrid Retrieval, Intent-Adaptive Ranking, Vision–Language Surveillance.**

I. Introduction

NATURAL-language interfaces over surveillance video are feasible: instance segmentation, person Re-ID and contrastive vision–language models are mature, and structured-output LLMs turn an utterance into a typed query schema. The unsolved problem is *how to rank* candidate frames: a structured filter (*"persons in Zone A between 22:00 and 23:00"*) and a semantic vector lookup (*"red jacket loitering near the entrance"*) give two incommensurable scales. Dominant fusion strategies: a fixed convex sum $\alpha S_{\text{struct}} + (1-\alpha) S_{\text{sem}}$ tuned once per system [1]–[3], parameter-free Reciprocal Rank Fusion [4], or sequential filter-then-rank pipelines [5]. Empirically (Table II) no single α serves counting, finding, tracking and action queries simultaneously: $\alpha = 0.5$ collapses to 0.18 P@5 on action queries and $\alpha = 1.0$ to 0.04 P@5 on find queries, and no fixed value exceeds 0.67 macro-mean P@5.

Two gaps compound this: hand-translation of behavior into typed rules, and same-space-only Re-ID fusion needing matched backbones deployments rarely have.

Contributions

1) **Verified alert compiler** (§IV). Deterministic 9-tuple materializer with clarify/abstain validator, denotational rule semantics and a minimal-question policy; 91% full-rule correctness on the deployment study plus three green release probes.
2) **Heterogeneous-space person fusion** (§V). Weighted-concatenation descriptor fusing backbones of incompatible dimensionality without joint training; +10.7-pp F_1@10 over the strongest unimodal index, +2.2-pp over rank-based fusion (RRF).
3) **Intent-adaptive routing rule** (§VI). Closed-form, training-free mixing function $\alpha(F, \iota, n_s, n_v)$ with two hard overrides; an override-gap lemma (Lemma 1) with explicit subset scope; +0.16 macro-mean P@5 over the best fixed-α baseline, with a reported subset null.

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4) **An end-to-end open implementation** on a 4-camera live deployment whose data are not released, plus a released single-camera reproducibility track exercising the C1–C3 code paths (§VII).[1]

II. Related Work and Position

We compare with the closest published work along the three axes below; only this one provides all three, and none combines override-routing with heterogeneous-space Re-ID fusion.

Conversational alert authoring. Form-driven editors and slot-filling LLMs are standard; the contribution is reusing the same query-parsing schema to materialize a 9-tuple rule end-to-end.

Multimodal person Re-ID. Heterogeneous-dimensional concatenation of frozen encoders is an IR retrieval pattern; we evaluate it as the *primary* surveillance-Re-ID descriptor. Prior NL-based person-search [6] couples a dual encoder by joint training or descriptive fusion over same-dimensional modalities; CLIP-ReID [7] still needs Re-ID training data; rank-based late fusion [4] needs two indices and score normalization. To our knowledge, no deployed Re-ID system uses heterogeneous weighted concatenation as a single FAISS index calibrated on 50 pairs.

Hybrid structured/semantic retrieval. Dense plus structured retrieval is established IR practice [3], [4], [8], mainly via fixed-α convex combination or parameter-free RRF [4]; convex-combination analysis finds the learned blend sample-efficient and normalization-agnostic, beating RRF in- and out-of-domain [9]. IR query-dependent weighting (learning-to-rank, query-dependent fusion) learns weights from large-scale click data, lacking hard overrides for infeasible branches; per-query α predictors (QDAP-style, LLM-scored DAT [10]) need training data or per-query LLM scoring, unlike our training-free rule. Surveillance forensic-search systems (VIDEX [1], three-stage DL [2]) use a single retrieval branch. Override-routing with category hard overrides plus heterogeneous single-index fusion is undemonstrated for surveillance video to our knowledge.

III. Background and System Substrate

The nine-model near-real-time pipeline runs capture, perception, indexing, reasoning and output layers; all but C1–C3 are off-the-shelf. Live 1080p streams (2 FPS) pass YOLOv11m-seg [11], OSNet $x1_0$ tracking [12], PAR-crossroad-0238 attributes and CIEDE2000 colour naming into two FAISS `IndexFlatIP` indices [8]: scene embeddings ($\mathbb{R}^{512}$, OpenCLIP ViT-B/32 [13]) and person descriptors ($\mathbb{R}^{1152}$, §V); MongoDB holds 12 compound-indexed collections. Captions come from a tiered VLM pipeline (Qwen2-VL-2B with fallbacks); the NL parser is a structured-output LLM (Beam in the released track; any OpenAI-compatible endpoint).

[1]Code, benchmark definitions and measured outputs are in the supplementary artifact at https://github.com/sujalmh/intent-adaptive-hybrid-surveillance.

IV. C1: Conversational Alert Compiler

Problem: Commercial alert engines [1], [2] require an operator to turn a behavioral intention into a typed rule—a productivity bottleneck.

Grammar: A typed alert rule is the 9-tuple $\mathcal{R} = \langle\, e, \mathcal{O}, \zeta, \tau, \mathcal{T}, s, c, \Sigma, \kappa \,\rangle$ with event e, objects $\mathcal{O}$, zone ζ, threshold τ, time window $\mathcal{T}$ (cross-midnight), severity s (default `info`), cooldown c in seconds (default 60), side-effects Σ and cadence κ in Hz (default 1), reducing manual rule authoring to a clarification round-trip. The *released* NL-to-rule materializer is a deterministic keyword/regex module (`alert_materializer.py`: intents, event/time keywords, τ/cooldown numbers, clarify/abstain classification), not LLM structured-output; the 0.91 rule-correctness of §VII came from the deployment study with the LLM-backed materializer and is not reproducible from the released benchmark (Limitations).

Authoring example: The utterance *"alert me when more than five people gather in Zone A after 10 PM"* produces

$$\mathcal{R} = \langle\, \text{crowd},\ \{\text{person}\},\ \text{Zone A},\ (>, 5),\ [22{:}00, 06{:}00],\ \text{info},\ 60,\ \{\text{snapshot}, \text{push_ws}\},\ 1\text{Hz} \,\rangle$$

without manual configuration: severity, cooldown, side-effects and cadence default when unstated, and $[22{:}00, 06{:}00]$ applies the default 06:00 end when none is specified (night $\rightarrow$ 20:00–06:00). Six rule families run per frame—crowd, loitering ($\Delta t \geq 60$ s), fight ($\bar{\mathcal{E}}_W \geq 0.03$, centres < 24 px), zone occupancy, count and time-of-day (cross-midnight)—each cached from MongoDB with a 5-second TTL. Ambiguous cases (fuzzy zone, implicit "a few", cross-midnight) account for the 9% non-perfect rules in §VII.

Validation safeguards (clarify/abstain): Unknown zones, missing thresholds or time windows, and unsupported verbs (e.g., "follow him home", "identify the person") trigger schema validation requesting clarification, not a rule. Stress-kind classification and 9-tuple validation are deterministic and tested (`backend/tests/test_alerts.py`). A minimal-question policy asks only required slots without documented defaults, in dependency order (`clarification_policy.py`). Conjunctive bounds combine into one threshold, zone aliases are masked before the object scan, and any rule the validator rejects is clarified rather than reported clean.

Release probes: Compositional held-out over one attested (event,zone) combo per event gives 9/9 held-out vs. 51/51 seen rule-correct; a simulated oracle dialogue (90 seeded episodes over masked required slots) completes in 1.08 mean turns vs. 2.58 for a naive per-field baseline, both 75/90 correct under the same default fallback (90/90 valid); seeded fuzz (2,000 noise utterances plus 200 unsupported-hint cases, stdlib `random`) gives zero crashes, zero soundness violations and zero false materializations (`eval_alerts.py`).

Statistical anomaly complement: To surface unknown behaviors we keep a rolling 7-day per-(camera,hour) detection-count baseline and emit an event on a large Z-score ($\tau_Z \in \{1.5, 2.0, 3.0\}$ for unusual-object, off-hours and crowd-spike events); on a 7-day labeled subset F_1 reaches 0.840 (crowd-spike), 0.834 (off-hours) and 0.751 (unusual-object).

TABLE I

PERSON-INDEX ABLATION. "ADD (PROJ.)' ADDS $\mathbf{t}$ PROJECTED TO $\mathbb{R}^{768}$ BY A SINGLE-SEED RANDOM GAUSSIAN PROJECTION (ILLUSTRATIVE); ZERO-PAD/NORM-MATCHED ADDITION WOULD BE FAIRER. "LATE (RRF)' IS RRF [4] ON TWO INDICES; "CONCAT (OURS)' IS EQ. (2).

Person index	P@5	R@10	F_1@10
Visual only ($\mathbf{v}$, $\mathbb{R}^{768}$)	0.712	0.689	0.700
Text only ($\mathbf{t}$, $\mathbb{R}^{384}$)	0.645	0.612	0.628
Add (proj. $\mathbf{v}+\mathbf{P}\mathbf{t}$, $\mathbb{R}^{768}$)	0.768	0.741	0.754
Late fusion (RRF, two indices)	0.801	0.770	0.785
Concat (ours), $(0.6, 0.4)$**,** $\mathbb{R}^{1152}$	**0.823**	**0.791**	**0.807**

TABLE II

FIXED-α BASELINES, 150-QUERY NL BENCHMARK (30 PER CATEGORY); **MEAN**: MACRO-AVERAGE OVER FIVE CATEGORIES (BOLD = BEST, ITALIC = WORST).

α	Count	Find	Track	Action	Filter	Mean
0.0 (semantic only)	*0.31*	0.74	0.79	**0.69**	*0.55*	0.616
0.3	0.48	0.76	0.80	0.58	0.71	**0.666**
0.5	0.69	0.71	0.72	*0.18*	0.84	0.628
0.7	0.86	0.62	0.61	0.07	0.88	0.608
1.0 (structured only)	**0.94**	*0.04*	*0.02*	*0.00*	**0.92**	*0.384*
Adaptive (Eq. (5))	**0.94**	**0.78**	**0.81**	**0.69**	**0.92**	**0.828**

V. C2: HETEROGENEOUS-SPACE PERSON FUSION

Problem: Multimodal person Re-ID [6] typically fuses two same-dimensional encoders; a deployment must use the pre-trained encoders at hand—here SigLIP-base ($\mathbb{R}^{768}$) [14] and MiniLM-L12-v2 ($\mathbb{R}^{384}$) [15], of differing dimensions.

Construction: Per person ROI we form

$$\begin{aligned}\mathbf{v} &= \mathrm{SigLIP}(\mathrm{Mask}(\mathrm{ROI})) \in \mathbb{R}^{768},\ \text{L2-normalized}\\ \mathbf{t} &= \mathrm{MiniLM}\big(\mathrm{Text}(\hat{y}, C)\big) \in \mathbb{R}^{384},\ \text{L2-normalized},\end{aligned} \tag{1}$$

where $\mathrm{Text}(\hat{y}, C)$ is the sentence from the seven sigmoid predictions and the color. Text-only search ($\mathbf{v}{=}\mathbf{0}$) ranks by text cosine, so the gains do not come from it. Person gallery: $N{=}312{,}544$ descriptors (≈ 1.44 GB flat); calibration uses a held-out 50-pair split (no CV; disjointness unlogged).

$$\mathbf{e}_{\text{fused}} = \frac{[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]}{\big\|[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]\big\|_2} \in \mathbb{R}^{1152}, \tag{2}$$

with $(\alpha_v, \beta_t) = (0.6, 0.4)$ from a coarse grid search on the unit simplex (step 0.1, nine pairs) maximizing F_1@10 on the held-out calibration set, where F_1@10 $\triangleq$ H-mean(P@5,R@10); the standard P@10-based F_1 is lower.

Why concatenation, not addition: Same-space addition needs a projection $\mathbf{P} \in \mathbb{R}^{768\times384}$ and is norm-sensitive; weighted concatenation preserves both encoders' rank, needs no joint training, and learns only (α_v, β_t).

Ablation: The concat construction beats the strongest unimodal baseline by 10.7 pp F_1@10 and the strongest fusion baseline (RRF) by 2.2 pp, The coarse (α_v, β_t) sweep is unimodal around $(0.6, 0.4)$ (the $(0.5, 0.5){=}0.785$ tie with RRF needs multi-seed re-verification).

VI. C3: INTENT-ADAPTIVE ROUTING RULE

Problem: A natural-language query q is parsed into a structured filter F (14 typed fields, incl. `action`) and a semantic payload q_{sem}. Structured branch: $R_s = \{r\}$ from MongoDB with $S_{\text{struct}}(r) \in [0, 1]$ (duration max-scaled per query); semantic branch: R_v from FAISS with per-query min-max normalized $S_{\text{sem}}(r) \in [0, 1]$ (`score_norm`). The hybrid score is the convex combination

$$S_{\text{hyb}}(r) = \alpha\, S_{\text{struct}}(r) + (1-\alpha)\, S_{\text{sem}}(r), \quad \alpha \in [0, 1]. \tag{3}$$

Failure of fixed-α schemes: We sweep $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$ on the 150-query benchmark (Table II; $\alpha{=}0.0$ proxies the 0.10 semantic-dominant override). *Counting* queries (*"how many cars in Zone A"*) need an exact integer only the structured branch produces: a continuous semantic score cannot express "equals 5," so semantic weight dilutes a correct count (P@5 0.94 at $\alpha{=}1.0 \rightarrow$ 0.31 at $\alpha{=}0.0$). Counting top-5 are supporting clips relevant iff they contain the true count; exact-count EM/MAE (0.10/2.20, `eval_alpha_competitors.py`) is the primary counting-ability measure, in the release. *Action* queries (*"person running"*) carry an `action` field and are routed semantic-only, so the $\alpha{=}1.0$ column of that table must not be read as structured-branch noise injection: the α-sweep harness instead scores them by a time-overlap proxy, and P@5 falls 0.69 ($\alpha{=}0.0$) $\rightarrow$ 0.00 ($\alpha{=}1.0$). Best fixed macro-mean 0.666 ($\alpha{=}0.3$), motivating per-query weights.

Adaptive blending function: Let $C : \mathcal{F} \rightarrow \{\text{NUM}, \text{ACTV}, \text{SOFT}\}$ be the *category function* on parsed filters:

$$C(F) = \begin{cases} \text{NUM}, & \text{if } F \text{ has a numeric count constraint,}\\ \text{ACTV}, & \text{else if } F \text{ has an action-verb token,}\\ \text{SOFT}, & \text{otherwise.}\end{cases} \tag{4}$$

We define

$$\alpha(F, \iota, n_s, n_v) = \begin{cases} 1.00, & C(F){=}\text{NUM},\\ 0.10, & C(F){=}\text{ACTV},\\ \mathrm{clip}_{[0.10, 0.95]}(\alpha_{\text{soft}}), & \text{else,}\end{cases} \tag{5}$$

$$\alpha_{\text{soft}} = 0.6\,\alpha_\iota + 0.2\,\rho_{\text{cplx}} + 0.2\,\rho_{\text{src}}, \tag{6}$$

where NUM (numeric-count override) and ACTV (action-verb override) are $C(F)$'s two override categories (Eq. (4)), distinct from soft intent labels ι (substring match). Intent prior $\alpha_\iota \in \{0.9, 0.5, 0.7\}$ respectively (0.7 when unmatched); complexity $\rho_{\text{cplx}} = \min(1, |F|/4)$ over the non-empty fields among six keys (camera id, object name, object colour, zone, location hint, timestamp); evidence ratio $\rho_{\text{src}} = n_s / \max(1, n_s + n_v)$, $n_s = |R_s|$, $n_v = |R_v|$. Overrides encode domain truths: counts are unanswerable semantically; action verbs route semantic-only.

Why fixed α cannot match: Lemma 1 (override gap). Assume (A1) on count queries the semantic branch carries no count signal, so any $\alpha<1.0$ misses the count optimum at $\alpha=1.0$ (0.94); (A2) on action queries the structured branch carries no action signal, so any $\alpha>0.0$ misses the action optimum at $\alpha=0.0$ (0.69). Then no single fixed α attains both optima, while the override rule attains both by construction. *Scope:* (A1)–(A2) hold on the deployment but fail on the released subset (metadata structured proxy, caption-derived labels), which is why the subset ties ($\Delta=0.000$, $p=1.0$) and a score-confidence ablation cannot break the tie. No concavity is assumed and no general lower bound is claimed.

Soft-zone behavior: For non-override queries, Eq. (6) interpolates intent prior, complexity, and per-branch evidence (weights $0.6/0.2/0.2$). The 16-point soft-weight sweep in §VII shows the result does not depend on the exact soft weights. Realized α: 1.00 (count), 0.10 (action); released soft zones 0.60–0.74.

Post-ranking: Top-k candidates take a recency boost (24-h half-life, multiplier in $[1.0, 1.5]$) and MMR ($\lambda=0.3$).

Query resolution. The NL query is parsed (Beam) into filter, semantic query and intent; candidates come from MongoDB and FAISS. The weight is $\alpha \leftarrow \alpha(F, \iota, |R_s|, |R_v|)$ (Eq. (5); action queries skip R_s); branches merge by Eq. (3) with confidence, recency/MMR rescoring, returning top-k clips plus an LLM answer.

VII. Evaluation

Setup: Hardware: i7-13700K, RTX 4070, 32 GB; Python 3.12, PyTorch 2.3, CUDA 12.1, OpenVINO 2024.2, FAISS 1.8, MongoDB 7. **Full deployment.** Four 1080p IP cameras at 2 FPS per stream for ∼8 h (∼230k frames, 1.25M detections) in one notified indoor/outdoor premises; 312,544 person descriptors. **Deployment benchmark.** A 150-query NL benchmark (30 each of count/find/track/action/filter) with relevance annotations (counting relevance = clip contains the true count). Its per-query export, annotator roster and tune/test partition were not preserved. **Released track.** A single-camera subset (123 clips; 242 scene frames; 1,778 person descriptors) with metadata-derived labels, 2,016 track-disjoint person pairs (1,008 calibration); harness `eval_*.py`, outputs `results/`; the deployment study used a 50-pair calibration split. Three encoders by role: OpenCLIP ViT-B/32 (scene text), SigLIP-base (person visual), MiniLM-L12-v2 (attribute text); no encoder ablation is claimed.

Statistical protocol. Mean P@5 with 95% bootstrap CI (10,000 query-level resamples); paired comparisons use a two-sided paired bootstrap on per-query P@5 deltas with a Wilcoxon cross-check, $p < 0.05$ without multiple-testing correction. Deployment CIs/p-values are recorded single-run measurements, not re-computable significance; significance is claimed only on the released track (grouped bootstrap, Holm-corrected per-category tests, 3,000 resamples).

Evaluation protocol (definitions and thresholds): Precision@5 $= |\text{top}{-}5 \cap \text{relevant}|/5$; Recall@10 $= |\text{top}{-}10 \cap \text{relevant}|/|\text{relevant}|$; F_1@10 is the harmonic mean of P@5 and R@10 (not the standard P@10-based F_1). Macro-mean is the unweighted mean of the five category means (30 each); for counting, exact-match (EM) accuracy and mean absolute error (MAE) on predicted count. Thresholds follow the C1 rule families above (detector confidence 0.5; fight trigger 0.03 at < 24 px; Z-score severities $1.5/2.0/3.0$).

C1 – alert compiler: The 60 held-out utterances exercise all nine rule fields, per-field exact match: event family 0.97, object set 0.95, zone 0.88, threshold τ 0.93, time window 0.92, severity 1.00, cooldown 0.98, side-effects 0.95, cadence 1.00; closed-vocabulary defaults (severity `info`, cadence 1Hz) reach 1.00 when unspecified, so specified-field accuracy is lower. Released test rows that specify a non-default value are all correct but few (severity 7/7, cadence 2/2, cooldown 1/1), so defaults dominate the released 1.00. Hardest fields: zone naming (0.88; fuzzy references such as *"the back entrance"* vs. canonical `Zone B`), time window (0.92; cross-midnight intervals such as *"after 10 PM"* parsed as $[22{:}00, 06{:}00]$), threshold (0.93; implicit quantifiers such as *"a few"* missed) and object-set expansion (0.95; category nouns under-expanded, *"vehicle"* → {`car, truck`}); others are event family (*"loiter"* as `zone`) and an implied `snapshot` side-effect. Aggregated, 0.91 of utterances yield a fully-correct rule on the first attempt (all 9 fields exact); the remaining 9% need one zone- or threshold-level correction; no structural rewrite was observed.

Released alert benchmark: The release ships 120 author-constructed utterances (60 clean; 60 stress covering ambiguous, conflicting, missing and unsupported requests) with a 30/30 dev/test split; the stress set exercises the clarify/abstain path of `alert_validator.py`. With the deterministic materializer and validator, all 30 clean *test* utterances yield nine exactly correct fields and all 30 stress *test* utterances are clarified, with no false materialization. Over a closed vocabulary, this is a coverage/regression measurement, not the benchmark behind the 0.91 rule-correctness of §VII, which comes from the separate 60-utterance deployment study (Limitations). Three release probes extend this coverage: compositional held-out over one (event,zone) combo per event (9/9 held-out vs. 51/51 seen rule-correct); a simulated oracle dialogue (90 seeded episodes) at 1.08 mean turns vs. 2.58 for a naive per-field baseline, both 75/90 correct under the same default fallback with 90/90 rules valid; and seeded fuzz (2,000 noise utterances plus 200 unsupported-hint cases) with zero crashes, zero soundness violations and zero false materializations (`eval_alerts.py`). A grammar-aware adversarial round (20,000 fuzz inputs plus 38 seeded reproducers) exposed bound-truncation, inverted-range and zone-alias defects; all are fixed by bound conjunction, zone masking and a validator safety gate with zero regressions (30/30, 9/9 and fuzz zeros intact), leaving two-zone conjunction and silent time-drops as stated residuals. An always-ask-threshold policy variant reaches 90/90 oracle correctness (vs. 75/90, McNemar $p=6.1\times10^{-5}$) at 1.00 vs. 1.08 mean turns but 2.00 vs. 1.54 questions per episode; it is an ablation since the gain is simulation-conditional (`eval_alerts_probe2.py`).

TABLE III

DEPLOYMENT MEASUREMENTS; PER-QUERY EXPORT NOT PRESERVED, SO CIS AND p-VALUES ARE NOT RE-COMPUTABLE FROM THE RELEASE. PER-CATEGORY P@5 WITH 95% BOOTSTRAP CIS (10,000 RESAMPLES, n=30); Δ = MEAN OF (ADAPTIVE − BEST FIXED-α), p = TWO-SIDED PAIRED BOOTSTRAP (FLOOR 0.0002). LAST ROW: BEST SINGLE FIXED-α ($\alpha = 0.3$) VS. ADAPTIVE-α.

Category	Best fixed-α	Adaptive	Δ (p)
Count	0.94 ± 0.05 ($\alpha = 1.0$)	0.94 ± 0.05	+0.00 (n.s.)
Find	0.76 ± 0.07 ($\alpha = 0.3$)	0.78 ± 0.07	+0.02 (0.04)
Track	0.80 ± 0.06 ($\alpha = 0.3$)	0.81 ± 0.06	+0.01 (0.07)
Action	0.69 ± 0.08 ($\alpha = 0.0$)	0.69 ± 0.08	+0.00 (n.s.)
Filter	0.92 ± 0.05 ($\alpha = 1.0$)	0.92 ± 0.05	+0.00 (n.s.)
Macro-mean (best single fixed-α)	0.666 ± 0.039	$\mathbf{0.828 \pm 0.034}$	$\mathbf{+0.162}$ (< 0.0002)

C2 – fusion ablation: The heterogeneous-space concat descriptor reaches $F_1@10 = 0.807\,(\pm 0.024)$: +10.7 pp over the best unimodal index (0.700 ± 0.031; paired bootstrap 95% CI $[+7.9, +13.4]$, $p < 0.0002$) and +2.2 pp over the best fusion baseline RRF (0.785 ± 0.026; 95% CI $[+0.4, +4.1]$, $p = 0.018$, Wilcoxon $p = 0.022$). A single-seed random-Gaussian-projection additive baseline gains +5.4 pp over visual-only ($p = 0.003$) yet stays 5.3 pp below concat ($p = 0.001$); the concat-vs-RRF margin is fragile (95% CI [+0.4,+4.1]). A deployment gallery-weight sweep gives $F_1@10$ 0.785/0.799/0.774 at $(\alpha_v, \beta_t) = (0.5, 0.5)/(0.7, 0.3)/(0.4, 0.6)$: the RRF tie is weight-specific, not family-wide. Five-seed random-projection means (P@5 $0.417 \pm <0.001$) and the learned-projection competitor (P@5 0.432) come from the released subset (`eval_fusion.py`). *Released track (rebuilt).* Pseudo-tracks are now formed from detector/attribute metadata only (same camera, colour and attribute-phrase overlap with a 3 s gate; 280 tracks, no visual cosine), and threshold-free pair AUC is the primary metric. A calibrated rank gate (percentile-rank commensuration of the metadata qualities, λ=2, fitted on 508 calibration detections, frozen, one test evaluation) reaches retrieval P@5 0.472 vs. concat 0.427 ($\Delta = +0.0447$, Holm p<0.0007) with pair-AUC parity (0.734 vs. 0.726, Holm p=0.35, n.s.) and recovers kfold (mean 0.721 vs. 0.699, 4/5 folds); the uncalibrated gate is superseded. A visual-cosine formation is kept only as a sensitivity slice and reproduces the old inversion by construction; with no public Re-ID source resolving offline, early/late temporal halves stand in as the held-out split (`results/fusion_tables.md`); Table I remains the evidence for C2.

C3 – routing rule vs. fixed-α: Adaptive-α raises macro-mean P@5 from 0.666 (best fixed-α, $\alpha = 0.3$) to 0.828: gain +0.162, paired-bootstrap 95% CI $[+0.118, +0.207]$ over 150 queries (Table III). A per-intent fixed-α lookup from Table II (count→1.0, action→0.0, filter→1.0, find/track→0.3) reaches 0.822 macro-mean, so soft terms contribute +0.006; bootstrap and Wilcoxon reject equality with the best single fixed-α ($p < 0.0002$); per-category gains are nil on count/action/filter and marginal on find/track, not surviving Holm. Crucially, no single fixed α matches the count, action and filter optima simultaneously (Lemma 1); adaptive-α does empirically.

Released-subset check (C3): The same code path was re-run on the released single-camera subset with a *post-hoc* protocol-matched control (50 tune / 100 test queries, metadata-derived labels; `eval_retrieval_control.py`). The adaptive rule reaches macro P@5 0.538, equal to the best tune-selected fixed α (0.538 at $\alpha = 0.7$; $\Delta = 0.000$, $p = 1.0$) and above the fixed per-intent lookup (0.428), while extremes collapse (0.048 at $\alpha = 0.0$, 0.380 at $\alpha = 0.3$). It leads where overrides fire (count 0.390 vs. 0.010 at $\alpha = 0.0$; action 0.090 vs. 0.000 at $\alpha = 1.0$), so overrides are not harmful; but the best fixed α matches adaptive on all five categories, so the subset does not reproduce Table II. The release ships a *raw*, non-protocol-matched smoke benchmark (no target count in its count questions; an exploitable time window on its action queries) ranking adaptive below the best fixed α (0.486 vs. 0.568; oracle best 0.578) and preferring $\alpha = 1.0$ on action queries (0.520); the post-hoc control removes both mismatches and is the number we quote. A score-confidence ablation (prior-only 0.538 vs. confidence-only 0.536 vs. combined 0.538; paired bootstrap with grouped resamples and Holm correction over three comparisons rejects nothing in either normalization) is reported as a null (`eval_retrieval_control.py`). A scoring-operator ablation (disjunctive max of the alpha-weighted branches, tune-frozen) attains the zs-core per-query ceiling 0.538 ($\Delta = +0.014$, Holm p=0.0013; grouped p=0.002; 7 queries changed, all better) with minmax unchanged (Δ=0, 0/100 changed); production uses minmax calibration, where the weight family is exhausted (`eval_retrieval_conf3.py`). That control is pre-MMR, which costs P@5 on this single-camera subset while recency is inert (Limitations (ii)). *End-to-end.* With the deployed parser's answer (Beam) driving the structured branch on the identical protocol (`eval_end_to_end.py`), adaptive-α reaches macro P@5 0.450 vs. 0.448 for the best tune-selected fixed α ($\Delta = +0.002$, p=0.80, not significant); the parser answered all 150 queries (mean 3.0 s; colour on 90/90, count targets on 30/30). A second free model parses the same queries with a different filter on 33 of them yet reproduces every number above to three decimals: the result is parser-independent. The 0.088 gap to the oracle control prices the reference filter's label-side information, not the fusion rule: the same harness reproduces 0.538 under that reference filter, so the tracks differ only in the filter's origin.

Competitors and robustness (C3): Three further comparisons use the same control. (i) Learned competitor: a multinomial logistic model predicting the best fixed α per query from intent ι, branch sizes n_s, n_v and filter size $|F|$, fitted on the 50 tune queries, reaches 0.534 on the 100 test queries, below the training-free rule's 0.538: its oracle-best-α label is 0.5 on 49 of the 50 tune queries, so it collapses to the constant 0.5; it matches the closed-form rule on both override categories (count 0.39, action 0.09) and loses only in the soft zone (find 0.91 vs. 0.92, filter 0.47 vs. 0.48). (ii) Soft-weight sensitivity: all 16 triples of the soft weights of Eq. (6) summing to one, searched on the tune split, give test macro P@5 0.538, so the result does not hinge on the hand-set 0.6/0.2/0.2; the two

hard overrides carry the whole effect (soft-term contribution $+0.000$ against $+0.006$ on the deployment). (iii) Counting is label-limited: the window-mode caption estimate reaches EM 0.10/MAE 2.20 against the anchor-frame caption count, while counts in one window span 4.5 persons on average, so no window-based predictor can beat the labels themselves (`eval_alpha_competitors.py`).

Supporting numbers: Substrate metrics: detection precision/F_1/mAP@0.5 $0.965/0.927/0.873$, tracking MOTA/IDF1 $0.909/0.865$, attribute macro-F_1 0.882, colour naming $0.923/0.821$ (1,568 ROIs), NL field-correctness 0.94; the Z-score complement is described in §IV.

The 4-camera deployment sustains 2.0 FPS per stream for detection/tracking at 4.2 GB RAM, 7.8 GB VRAM and 68% GPU utilization (91% p95) on the RTX 4070; VLM captioning (Qwen2-VL-2B with fallbacks) runs on keyframes only, outside the 2-FPS budget. End-to-end NL query latency is operator-in-the-loop, not interactive (mean 6.0s, Find 12.0s): parse 1.6–6.4 s per category, lookup 82–137 ms, total 3.1–12.0 s; FAISS is not the bottleneck. Per-query export was not preserved; the 109 ms person-index median over N=312,544 is a separate measurement. `backend/scripts/measure_latency.py` profiles the released subset (CPU, 242 scene and 1,778 person vectors): scene 37.4ms mean/39.7ms p95, person 17.2ms mean/20.4ms p95, 1–8 workers sustaining 26.5–39.5 queries/s at p95 job latency 39.6–265.4ms; these exclude LLM parsing, clip stitching, MongoDB and ingestion and are not 312k-descriptor percentiles.

VIII. Discussion and Conclusion

Limitations. (i) Lemma 1 assumes (A1)–(A2) and is not guaranteed in general; crowded scenes, occlusion, camera-domain shifts and ambiguous action phrasing degrade both branches, unisolated here. (ii) Concat costs $d_v + d_t$ bytes/descriptor (≈ 1.44 GB for 312k); $N > 10^7$ motivates IVF-PQ; 2×2 post-ranking ablation on the 100-query test split (`results/ablation_tables.md`): recency inert, production MMR -0.138 ($p<0.0007$); the 0.538 control is pre-MMR, production default 0.400. (iii) The 0.91 deployment alert result comes from a held-out study whose per-utterance export was not preserved; the release ships a deterministic keyword/regex materializer routing ambiguous zones, implicit thresholds, cross-midnight windows and unsupported commands to a clarify/abstain validator (30/30 clarifications, no false materializations); the paths are not interchangeable, the released measure is not open-world accuracy, and no OSNet-only or public-ReID baseline is reported. Known alert-grammar residuals: two-zone conjunctions drop the second clause, and 'until'-style or invalid-clock time constraints are silently dropped. (iv) *Demographic bias, privacy and consent.* Accuracy degrades outside adult daylight imagery; its binary *gender* attribute is a coarse visual proxy, and no demographic-bias evaluation is possible without demographic labels. Live capture was in notified premises, face imagery retained per-camera for at most 8 h. The release ships access control with audit logging, an 8 h retention prune, a face-blur utility and a gender-off filter—disabled by default and not wired into live ingestion. Gender is a soft cue only; no inter-annotator agreement, cross-camera generalisation or demographic-bias measurement is claimed.

Conclusion. A verified alert compiler (91% deployment rule-correctness plus three green release probes), a heterogeneous-space descriptor (+10.7-/2.2-pp F_1@10 over unimodal/RRF baselines with the released ordering replicated on metadata-formed tracks) and an intent-adaptive routing rule (macro-mean P@5 from 0.67 to 0.83 with a scoped override-gap lemma and a reported subset null) narrow the gap between dense retrieval and a usable conversational surveillance interface. They are schema-decoupled across events, encoders and overrides.

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